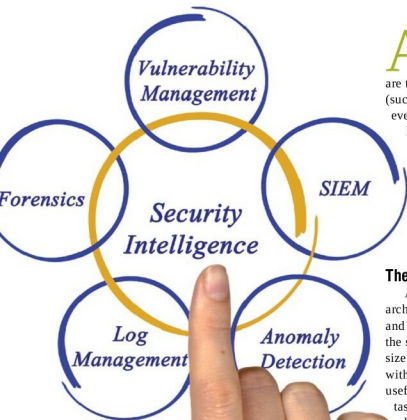


Setting up a Log Management System in a Data Centre

In a production environment, log management plays a key role. The sheer volume and diverse formats of the logs makes the job of studying them difficult without an LMS or log management system. Logstash can be used in conjunction with Kibana, Elasticsearch, Grok and various other tools, to create an LMS.



Any server, system or an application keeps a record of the significant events that happen across its various sub-components. These events are typically recorded with different significance levels (such as error, info, debug, etc). The files in which the events are collected are known as log files, and this logging may be centralised or distributed across the sub-components of the system. The log files are consulted by the operations personnel in the event of an error to identify the root cause of the problem and to fix it. The log entries typically contain information regarding the time of the event, component in which the event occurred and the description of the event.

The need for a log management system

A log management system (LMS) collates, stores, archives and provides utilities for searching log files and alerts administrators in case of critical errors in the system. As the log files tend to accrue and grow in size over a period of time, it is not uncommon to deal with log files of the order of Gigabytes. Extracting useful information from these log files is a complicated task not only due to the size, but also owing to the discrepancies in the format of log files generated by different sub-components of the system. Another complication is to tie up the log entries across multiple but related components that are involved in the root cause of the error. Typically, developers tend to make use of system utilities like *awk*, *tail* and *cat* along with *grep/regex* to look for clues regarding the errors. This approach is like searching for a needle in a haystack, and hence a more systematic and scalable mechanism is needed to tackle distributed and huge log files.

To summarise, the following common problems faced by the operations teams in organisations necessitates the use of an LMS:

- Diverse logging formats employed by different sub-systems. For instance, the timestamp format/time zones used by